

- ◆ I was responsible for:
 - ◆ This migration scenario (and these slights)
 - ◆ The hardware configurations of all the z196 mainframes in PLM and Windhof during the startup fase
 - ◆ The hardware configurations of all the z196 mainframes in PLM, Windhof and Kayl during the transition fase
 - ◆ The hardware configurations of all the z196 mainframes in Windhof and Kayl during the final fase
 - ◆ The hardware configurations of all the z196 Ficon Directors in PLM and Windhof during the startup fase
 - ◆ The hardware configurations of all the z196 Ficon Directors in PLM, Windhof and during the transition fase
 - ◆ The hardware configurations of all the z196 Ficon Directors in Windhof and Kayl during the final fase

- ◆ Deze informatie beschouw ik niet als bedrijfskritisch en strikt vertrouwelijk:
 - ◆ Dit scenario is meer dan 10 jaar oud, er wordt nu gewerkt met:
 - ◆ Recentere modellen van mainframes
 - ◆ Recentere versies van IBM Z Storage Suites
 - ◆ Er worden geen interne configuraties + migratiemethodes getoond
 - ◆ Dexia bestaat niet meer
 - ◆ IS4F bestaat niet meer
 - ◆ Een deel van deze oude informatie is nog steeds beschikbaar op het internet via andere bronnen



New SAN

Renewal of San Switch combined
with DC move from PLM to Kayl

Preferred scenario

1. Order + install Directors + Temporary Switches
2. Order + install Tape
3. Start migration Tape
4. Order + install DASD
5. Start migration DASD (cascading)
6. Order + install MF
7. Wait Swap MF until finish TAPE migration
8. Swap MF + DASD simultaneously PLEX by PLEX

ISL:

- Starting with 2x8 8Gbit shortwave (upgradable to 16Gbit links after 2016)
- Fiber length ADVA-ADVA between Windhof and Kayl: **24,2 and 23,6km**
- **PLM > LW-SM > Hitec > SW-MM > Kayl (tape)**
- Performance july 2014 / per director: avg 746MB/s ; max 1.565 MB/s

Infiniband :

- 2x8 5Gbit longwave(SM) connections

DISKS:

- DS8870: Still 8Gbps (16Gbps Q1 2015?)
- DS8800-2: 4Gbps
- DMW205: 46 x 2107 CU; DMW206: 30 x 2107 CU; DMW222: 12 x 2107 CU

VSM:

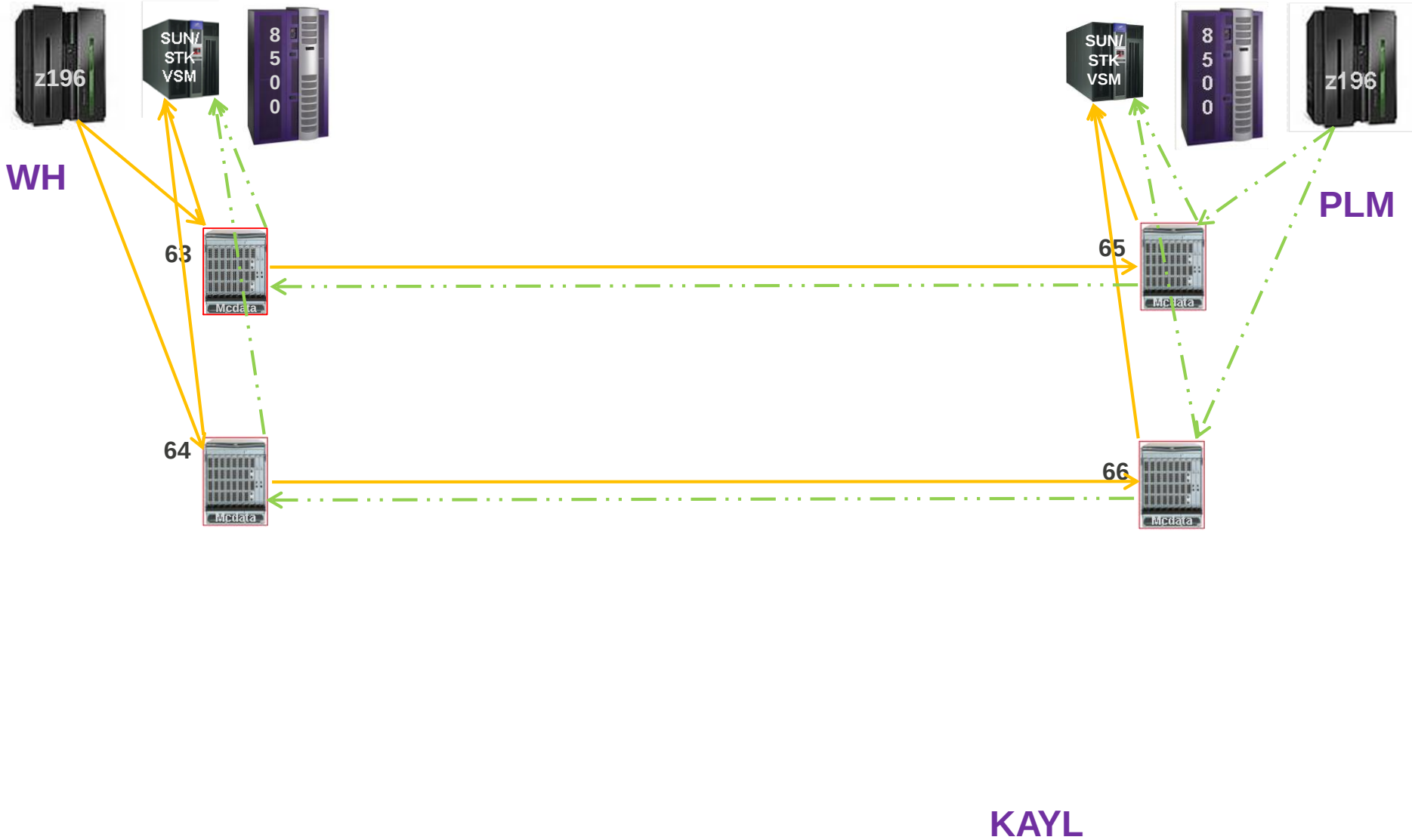
- 400TB compressed (non duplexed)
- Target: each site 2xTS7720 (or 1xTS7720+expansion frame)

Ficon Directors AS-IS:

- SMW103 avg: 2880 MB/s; Max: 6500 MB/s
- SMW204 avg: 2850 MB/s; Max: 6400 MB/s
- SMP105 avg: 1700 MB/s; Max: 3400 MB/s
- SMP206 avg: 1700 MB/s; Max: 3400 MB/s
- 110 ports/director in use

Ficon Directors Target:

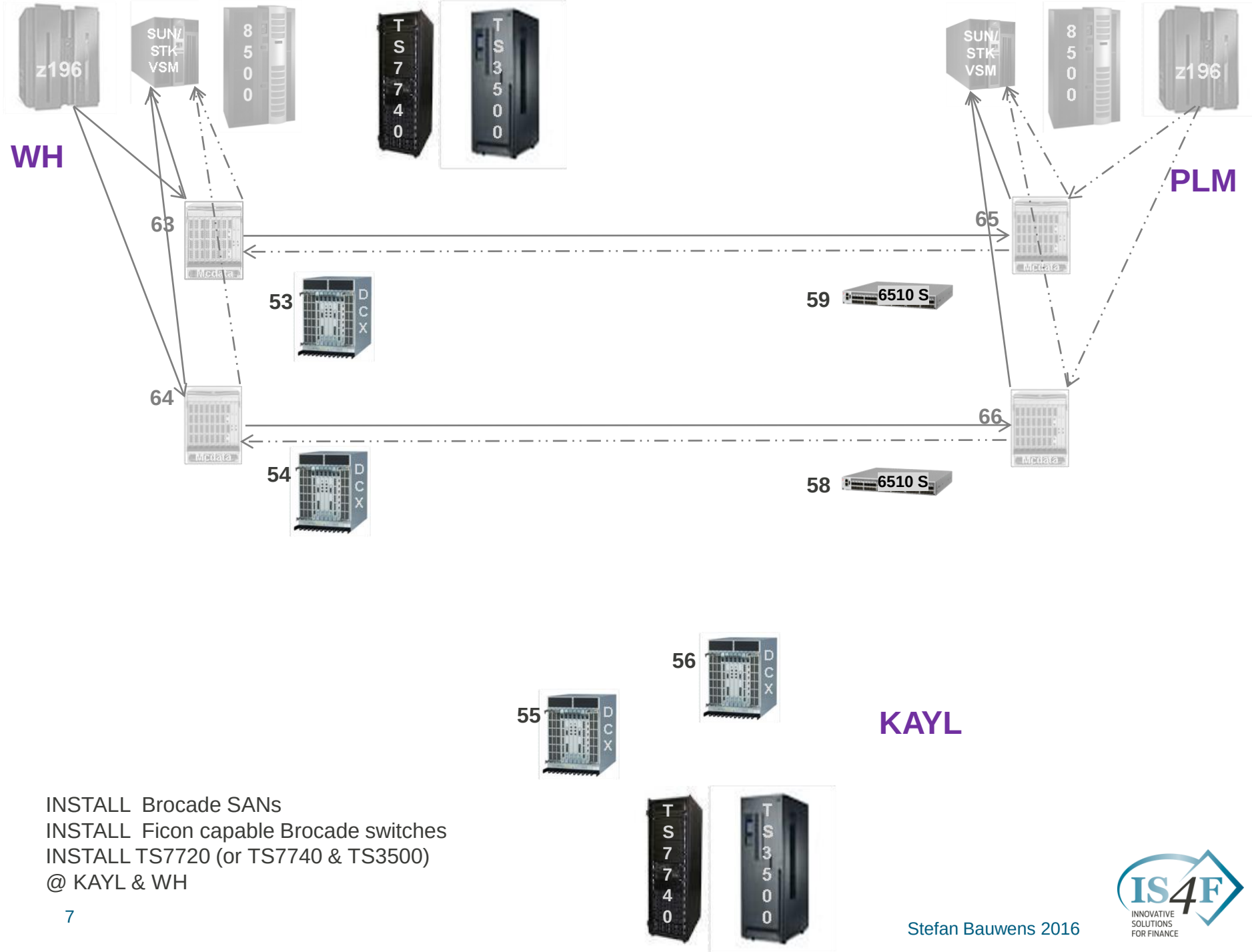
- 2x2xSAN384B-2
- SAN384B-2
 - 4x32x16Gbps port blades (128ports)
16Gbit + 8Gbit
 - 4 blades with 2xISL ports (or 2 blades for 4xISL trunking)
 - CUP
 - Local switching (ex. MFR1 > DS8870: x'00' + x'01')
 - npiv for AS400



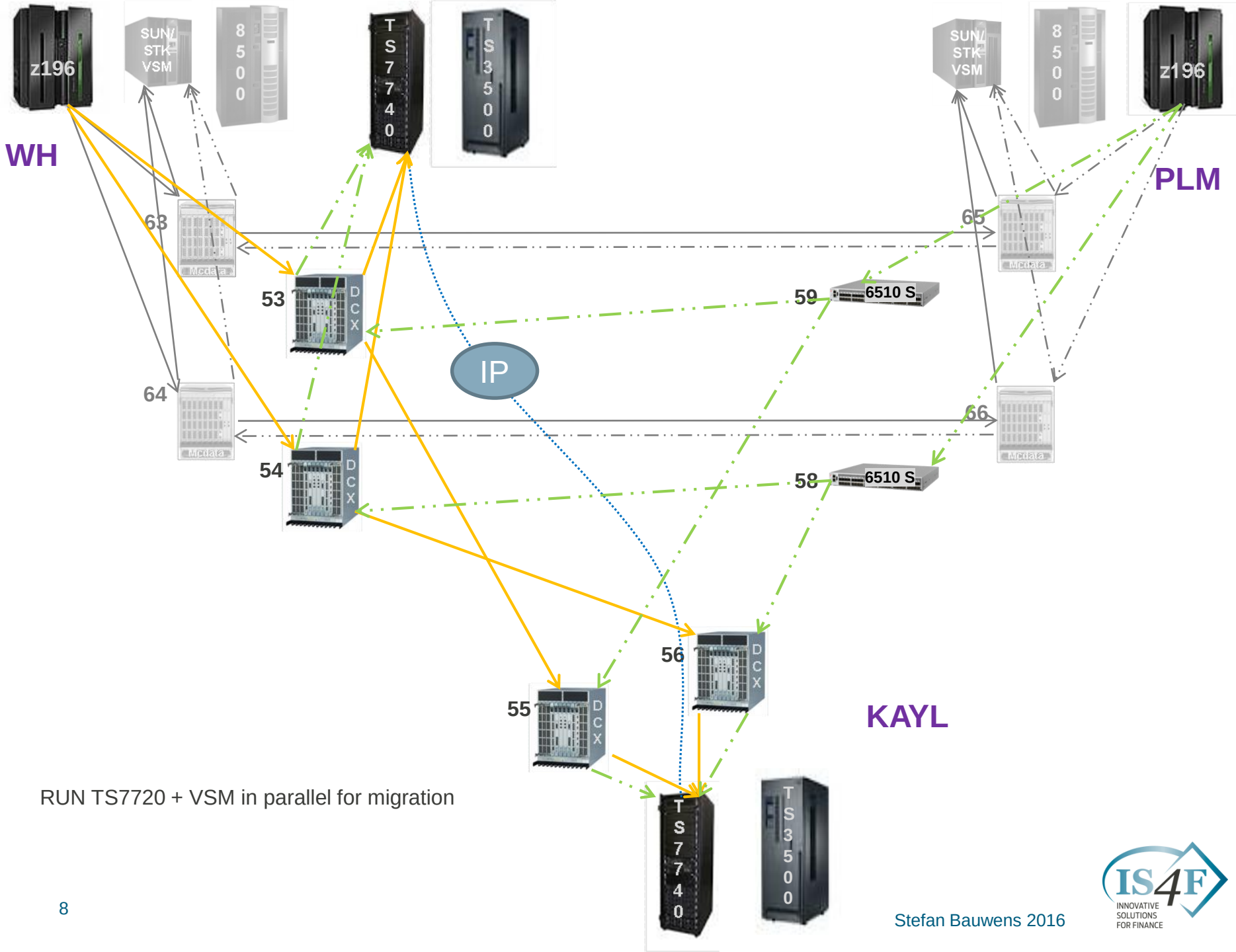
AS IS (TAPE)

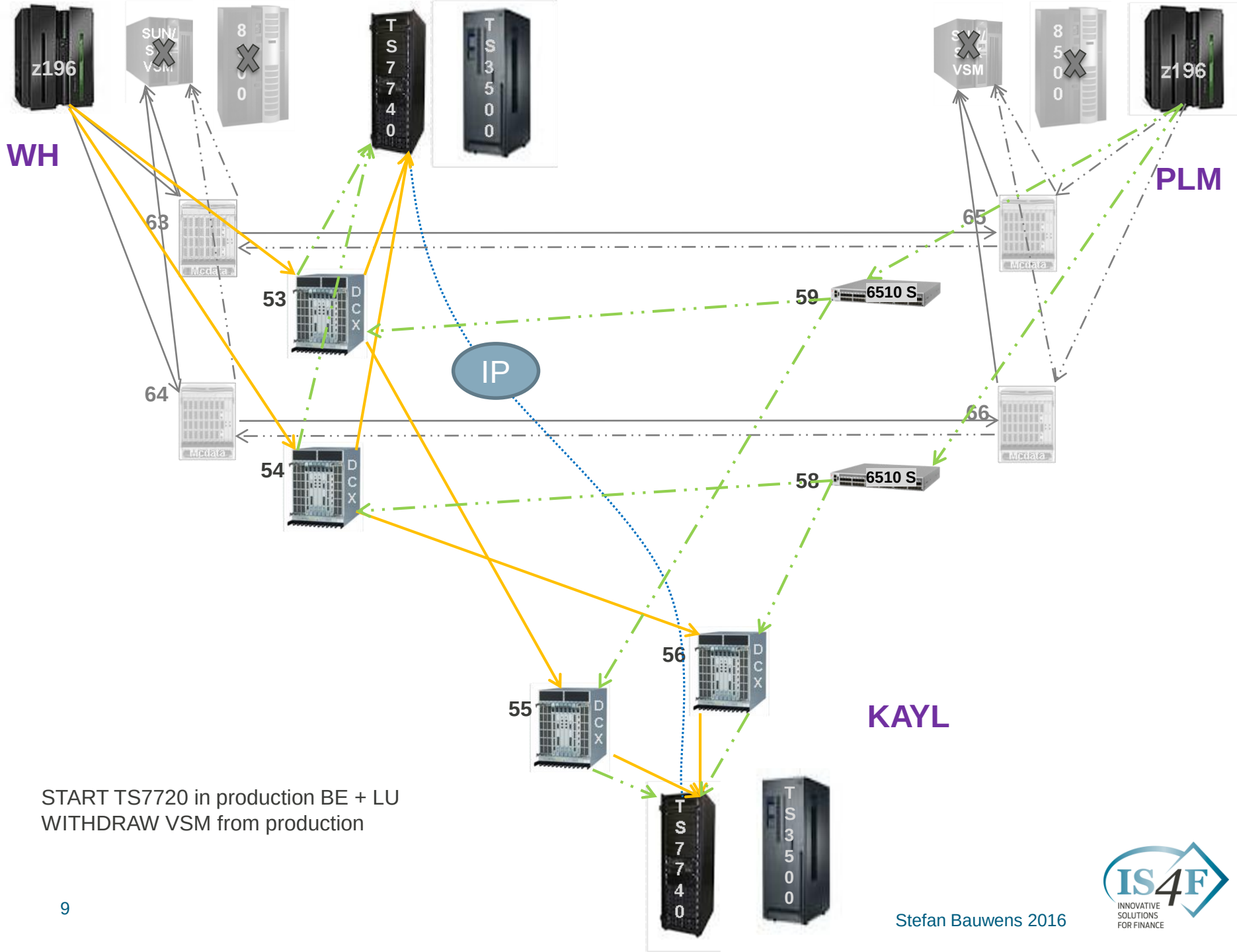
→ data path from MF1

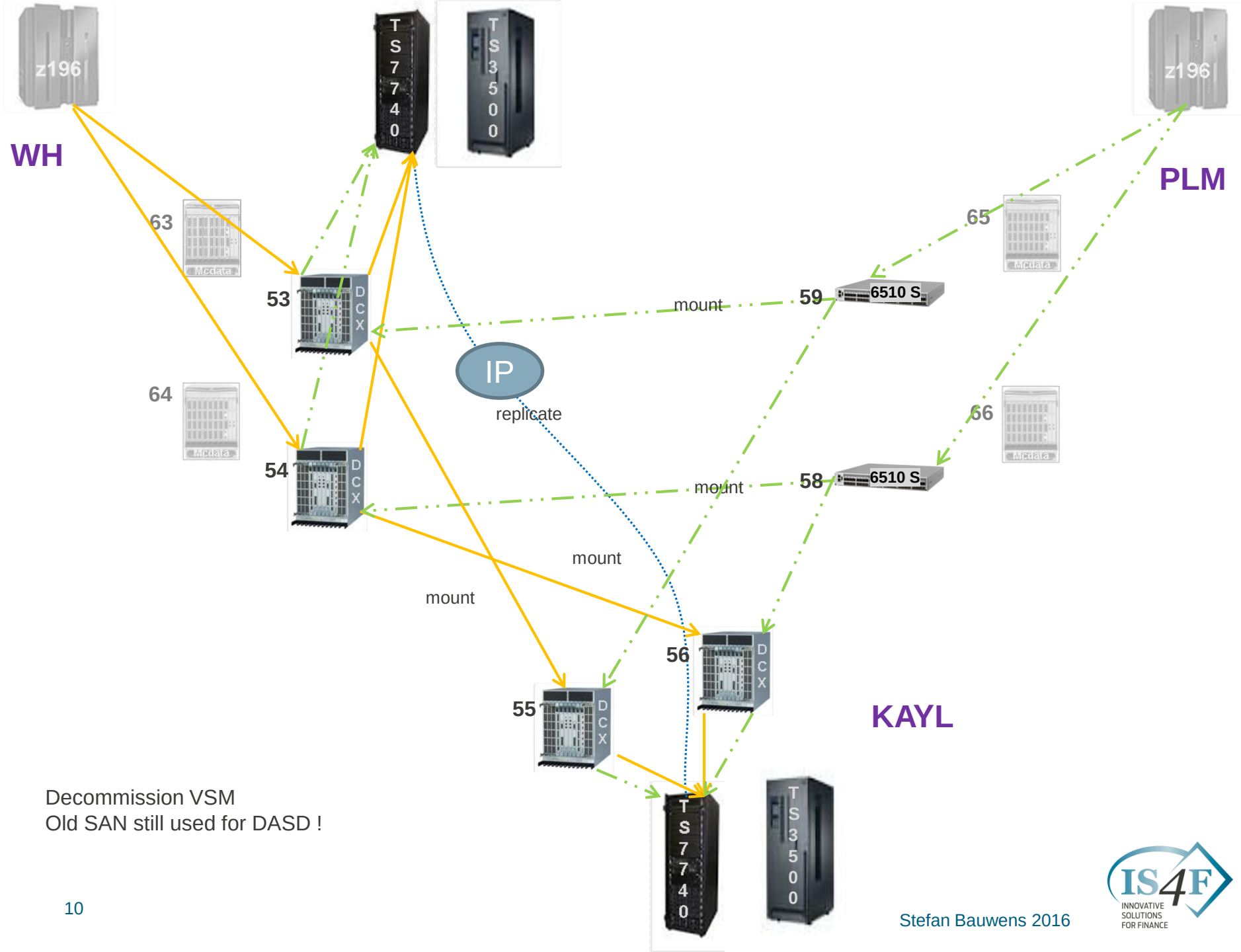
- - - - -> data path from MF2



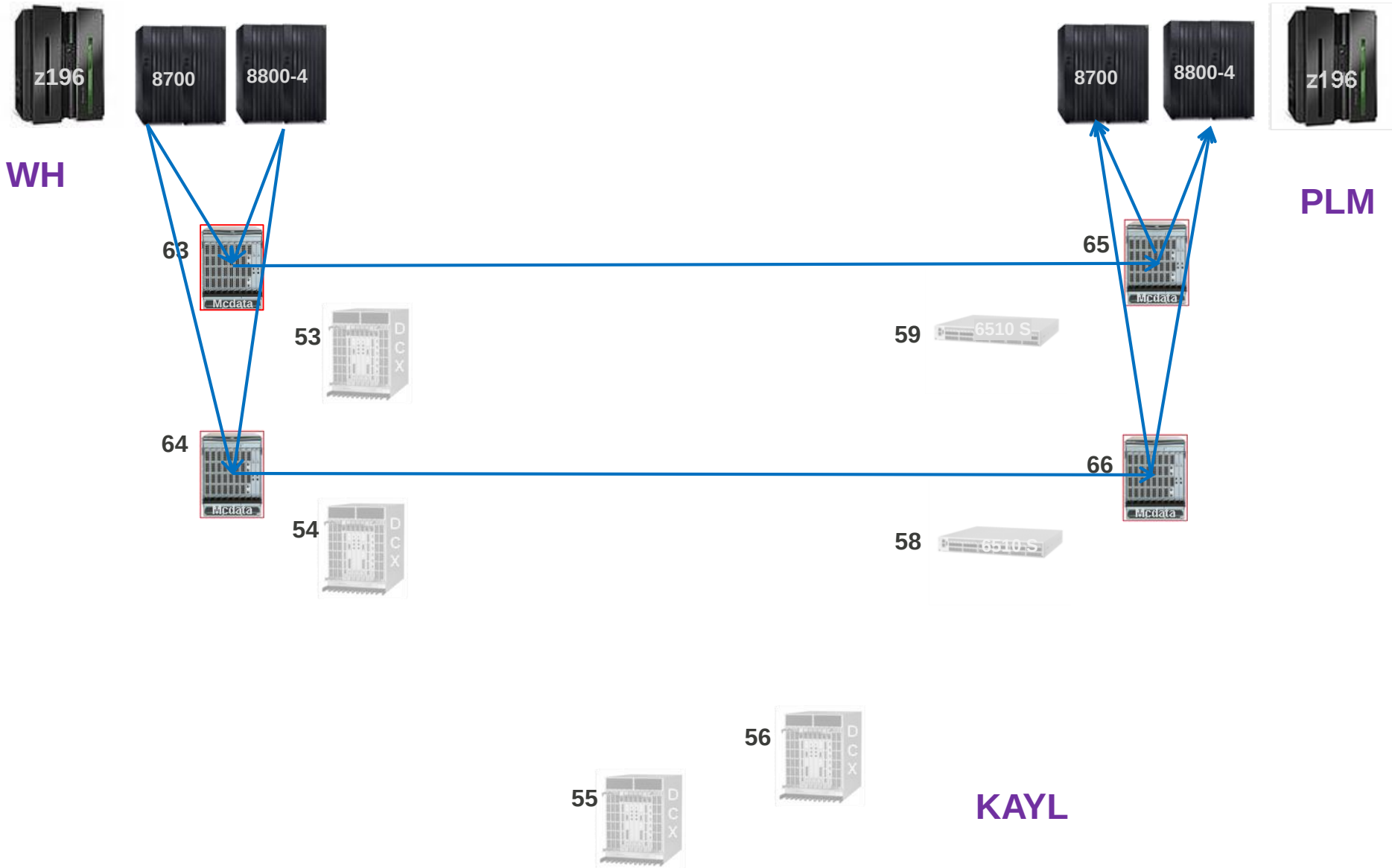
INSTALL Brocade SANs
 INSTALL Ficon capable Brocade switches
 INSTALL TS7720 (or TS7740 & TS3500)
 @ KAYL & WH







Decommission VSM
Old SAN still used for DASD !



AS IS (DASD)

→ pprc paths over old SAN

